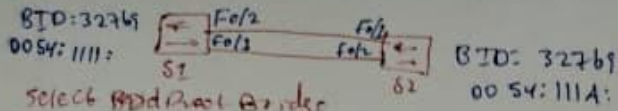


Client-ge marheli chano IPv4 saxali, jidax aqel waziran idhe Gagan.

Topology 1



Select Root Bridge

→ S1 → root bridge because has a lowest MAC address.

→ Root port

S2 → interface Fa/20, is root port because sender port ID is S1 Fa/20.

Designated Port

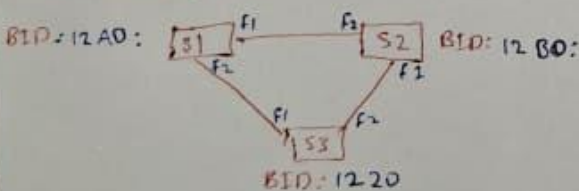
→ S1 → two interfaces are Designated port, because the switch become Root bridge Port are designated.

→ S2 → has not designated port

non designated port

→ S2 → interface Fa/24 has Alternate.

Topology 2



1) Root Bridge

S3 → is a Root Bridge because is the lowest MAC address

2) Root port

S1 → Fa/20, is a root port, because is a lowest Root Port Cost

S2 → Fa/20, is a root port because it is lowest Root Port Cost.

3) Designated port

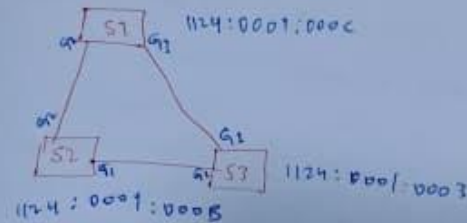
S3 → two interfaces are designated port.

S1 → Fa/20, interface Fa/20 is designated, because is lowest sender Bridge ID.

4) non-designated

S2 → Fa/24 is alternate.

Topology 3



1) Root Bridge

S3 → IS a Root Bridge, because it is the lowest MAC address.

2) Root Port

S1 → interface G2 is root port, because it is the lowest cost path that reach Root Bridge.

S2 → interface G1 is root port.

3) Designated port

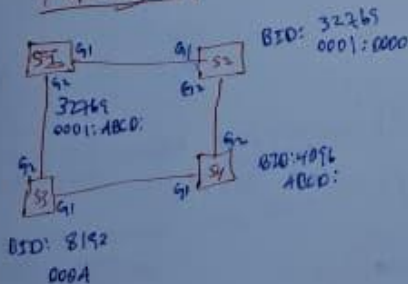
S3 → all interfaces are designated port.

S2 → interface G2 is a designated port.

4) Non Designated port

S1 → interface G2 is alternate port.

Topology 4



1) Root Bridge

S4 → IS a Root Bridge because it is the lowest MAC address priority.

2) Root port

S2 → G2 interface is a root port.

S3 → G1 interface is a root port.

S1 → G2 interface is a root port.

3) Designated port

S4 → all interfaces are designated.

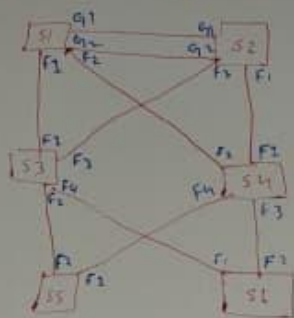
S3 → G2 is designated port.

S2 → G1 is designated.

4) Non-Designated

S1 - G2 is alternate port.

Topology



SW1 = 0001:45A3:3342
 SW2 = 0001:45A3:3322
 S3 = 0004:45A3:334B
 S4 = 0004:45A3:33DD
 S5 = 0A14:563C:0001
 S6 = 1234: A124:3371

STP Tie-Breaking Questions

Q.1

First Factor used to elect root bridge is BID, first priority, if priority are same, then MAC address.

Q2

a non-root switch use to select its root port,
 First lowest cost path reach root bridge.
 Second: lowest sender BID
 Third: lowest sender port ID
 Fourth: lowest local port ID.

Q3

If two possible paths have the same root path cost, checking or calculating lowest sender BID.

Q4

The purpose of designated port is every segment has one port that can forward frame.

Q5

If ports can't become root port or designated port is non-designated port. Non-designated port can't forward traffic.

Root Bridge

S2 → root bridge

Root port

S1 → G1
 S3 → F3
 S4 → F1
 S5 → F1
 S6 → F2

Designated port

S2 → G1, G2, F3, F4
 S1 → F1, F2
 S3 → F2, F4
 S4 → F4, F3

non-Designated

S1 → G2
 S3 → F1
 S4 → F2
 S5 → F2
 S6 → F1